Teaching Statement

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Andrea Salvanti

During my Ph.D. studies, I gained graduate-level teaching experience at the Barcelona School of Economics, spanning microeconomic theory, mathematics, and econometrics. I was the TA for Advanced Microeconomics II (Prof. Antonio Penta), covering general equilibrium and game theory; the Brush-up in Advanced Mathematics (Prof. Alexander Frug), covering the mathematical foundations needed for graduate economics; and Advanced Econometric Methods II (Prof. Christian Brownlees). These experiences have prepared me to teach across the core curriculum at the Ph.D. level, with particular strength in microeconomic theory and game theory. I am also prepared to teach Microeconomic Theory, Mathematics for Economists, and Econometrics at the undergraduate and master's levels.

I particularly enjoy teaching theoretical material because a well-understood framework gives students the tools to understand economic mechanisms and approach questions they have not encountered before. My teaching approach has three connected elements: establish the economic question and its theoretical foundations; make explicit the modeling choices and assumptions that connect the framework to that question; and build intuition through examples and applications. I then connect the framework to empirical evidence, either by discussing how experiments can be designed to test its predictions or by confronting those predictions with real-world data.

I place particular importance on the interpretation of theoretical frameworks through concrete questions, making sure that students understand which economic problems a framework is designed to tackle. I emphasize that what we can learn from a model depends both on modeling choices, such as the primitives we choose to include, and on the assumptions we impose on those objects and on behavior, such as the equilibrium notion we adopt to close the model. In game theory, for instance, I make explicit the primitives of the problem, such as players, feasible actions, and payoffs, and distinguish them from the behavioral assumptions associated with the equilibrium concept, emphasizing the distinct role played by each. This perspective also reflects my research in behavioral economics, where much of the intellectual work lies in formulating assumptions about behavior precisely and testing their implications. In econometrics, I similarly distinguish assumptions that determine the functional form of a structural equation, such as linearity, from those that identify its parameters, such as an appropriate exogeneity assumption on the error term. Together with the structural interpretation of the equation, these identifying assumptions allow the coefficients to be given a causal interpretation.

An important part of this perspective is connecting theoretical frameworks to empirical analysis while keeping identification distinct from estimation. Identification asks whether the object of interest is pinned down by the model and observable data under the maintained assumptions; estimation asks how to recover that object from a finite sample. A model therefore does not, by itself, dictate a unique estimation method. In game theory, for example, once the primitives of a game and an equilibrium concept are specified, the model may deliver sharp, sometimes unique, predictions about behavior; confronting those predictions with data is a separate empirical step. This may require an experimental design that generates the variation needed to test the theory, or structural estimation that recovers which strategies subjects use. Across theory and econometrics, I want students to distinguish clearly

between what a model predicts, which assumptions make those predictions sharp, and how data can be used to estimate or test them.

I place particular value on preparing lecture notes tailored to each course. My preferred format makes the question, notation, assumptions, and main argument easy to follow, with worked examples followed by exercises that gradually remove the guidance. Problem sets would then ask students to apply models to clearly defined economic questions and, where appropriate, take them to data. Beyond solving a specified model, I would ask students to propose the framework and relevant assumptions themselves and, when the exercise involves data, to state the identifying assumptions. By articulating the assumptions needed for a result and examining what changes when one is relaxed, students would learn to explain the role of each restriction rather than treat assumptions as a checklist. Where appropriate, assignments could ask students to evaluate an AI-generated explanation or piece of code, identify unstated assumptions, and correct errors. I would pair permitted AI-assisted work with in-class exercises or brief oral explanations to assess students' own understanding. The objective is for students to use these tools critically, without delegating the economic reasoning or responsibility for their conclusions.

Beyond the classroom, I would welcome supervising undergraduate and graduate research. I would help students turn a broad interest into a focused question, develop an appropriate theoretical framework, and identify evidence that could distinguish between competing explanations. I would not limit supervision to topics that overlap directly with my own research; as long as I can provide useful guidance, I would be happy to supervise students working across a broad range of topics, particularly when I can contribute to the formulation of the research question, the theoretical framework, or the empirical strategy.